PORTFOLIO CASE STUDY

Diagnosing Revenue Decline in a Mobile-First Commerce Ecosystem

Behavioral Analytics | Safari Mobile | Core Web Vitals | Search Visibility | Revenue Modeling

ROLE: DIGITAL DIRECTOR / PRODUCT & GROWTH STRATEGY

2024-2025

The diagnostic challenge

Revenue was declining across several digital channels at the same time, but the signals did not point to one obvious cause. Some conversion and profitability indicators were improving even as traffic and total sales contracted.

-77%

MARKETING CAPACITY

Recovery could not depend primarily on additional media spend.

MIXED

PERFORMANCE SIGNALS

Traffic, conversion, browser, search, and margin trends moved in different directions.

SILOED

REPORTING

Search, storefront, product, campaign, and financial evidence lacked one shared diagnostic frame.

The diagnostic method

1. **Segment** — Browser + device
2. **Validate** — Real-user UX
3. **Model** — Traffic vs conversion
4. **Prioritize** — Impact vs effort
5. **Translate** — Findings to action

CENTRAL QUESTION

Was revenue falling because the storefront was failing - or because too few qualified customers were reaching an increasingly effective experience?

CONFIDENTIALITY NOTE

All visuals are reconstructed from internal analysis and anonymized. Company identifiers, proprietary screenshots, customer-level data, campaign creative, and vendor materials are excluded. Modeled scenarios are labeled separately from actual performance.

Separating symptoms from root causes

Blended averages could support the wrong intervention. The analysis separated acquisition, experience, conversion, and revenue quality before assigning a cause or recovery action.

ACQUISITION	→	EXPERIENCE	→	CONVERSION	→	REVENUE QUALITY
Search visibility Social and referrals Product-feed presence Traffic volume + quality		Mobile behavior Browser-specific friction Core Web Vitals Landing-page relevance		Conversion rate Estimated orders Product and checkout behavior New vs returning users		Revenue volume AOV recovery opportunity Gross profit + margin Retention potential
Who reached the site?		**What happened after arrival?**		**Did qualified visitors act?**		**What value remained?**

EVIDENCE SOURCES

- Shopify analytics
- Browser/device segments
- Real-user monitoring
- Google Search signals
- Merchant/feed health
- Klaviyo audiences
- Channel performance
- Financial context

DECISION LOGIC

1. **Where did the decline originate?** Segment by browser, device, source, and customer type.
2. **Was storefront friction still primary?** Use real-user performance and conversion evidence, not assumption.
3. **Which lever held more recovery value?** Model traffic restoration separately from conversion improvement.
4. **What was recoverable under constraint?** Prioritize organic visibility, feed health, retention, and product relevance.

Lower revenue did not automatically mean the storefront was converting poorly.

Traffic loss overwhelmed conversion improvement

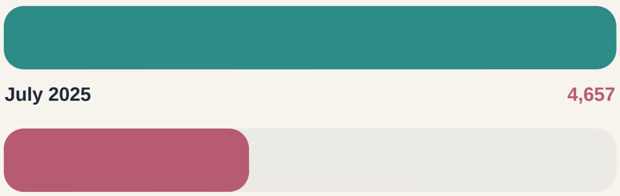
A year-over-year Safari Mobile comparison revealed that the conversion rate improved, but the available audience shrank too sharply for that gain to protect order volume or revenue.

ACTUAL PERFORMANCE

Safari Mobile sessions

July 2024 — 11,694

July 2025 — 4,657



-60.2% TRAFFIC

Conversion rate

0.97% JULY 2024 → **1.37%** JULY 2025

+41.2% RELATIVE

ESTIMATED ORDER EFFECT

~113 ORDERS

Estimated July 2024 Safari Mobile conversions.

~64 ORDERS

Estimated July 2025 Safari Mobile conversions.

-43.8%

Estimated order decline despite the stronger conversion rate.

CONSERVATIVE REVENUE RECOVERY MODEL

ACTUAL JULY 2025

$18.5K

Safari Mobile revenue

→

MODELED AT RESTORED TRAFFIC

$46.3K

Maintaining 1.37% conversion and ~$289 implied AOV

+$27.9K

+151%

DIAGNOSTIC VERDICT **Traffic and reach loss - not conversion collapse - was the larger constrained recovery opportunity.**

Testing whether the storefront remained the primary constraint

Real-user evidence showed a mobile-led audience and materially improved Core Web Vitals. That changed the diagnosis from “the site is still broken” to “the optimized experience is underfed by qualified traffic.”

REAL-USER DEVICE SAMPLE

7,688 mobile sessions · 4,622 desktop sessions · 120 tablet · 7 other

CORE WEB VITALS - REAL USER P75

LCP

~1.75s

Largest Contentful Paint

28% improvement

INP

64-72ms

Interaction to Next Paint

70% improvement

CLS

0.01

Cumulative Layout Shift

90% improvement

STATUS SHIFT

FAILING → FULLY GREEN

Performance work remained important, but the evidence no longer supported treating speed as the only or primary explanation for revenue decline.

WHAT THE EVIDENCE CHANGED

PRIOR ASSUMPTION

Lower revenue likely reflected a storefront that was still too slow or unstable.

VALIDATED FINDING

The optimized mobile experience performed within strong real-user thresholds.

RECOVERY IMPLICATION

Shift priority toward qualified traffic, product visibility, and landing relevance.

04 / RECOVERY PRIORITIZATION

Turning fragmented findings into low-cost, high-leverage action

With marketing spend reduced by 77%, the strategy emphasized recoverable structural issues: product visibility, feed health, organic discovery, owned-channel quality, and evidence-led mobile merchandising.

PRIORITIZATION MATRIX

RECOVERY WORKSTREAMS

1. **Search + feed health**

   Diagnose product-feed and Shopping visibility issues; strengthen product eligibility and discovery.

2. **Product architecture**

   Use structured attributes, collections, PDP language, and metadata to improve relevance.

3. **Owned audience quality**

   Separate total profiles from active subscribers; use suppression, engagement, and retention signals.

4. **Mobile-first merchandising**

   Prioritize Safari-heavy behavior, landing quality, hierarchy, and product relevance.

5. **Measurement cadence**

   Track channel, browser, conversion, margin, and recovery assumptions separately.

IMPLEMENTED VS. MODELED

Implemented: performance optimization, product and content architecture, browser/device analysis, search and feed diagnosis, lifecycle analysis. Modeled or recommended: restored-traffic revenue scenario, broader search recovery, expanded low-cost acquisition and retention roadmap.

PRIORITIZATION PRINCIPLE

Repair the structural paths that qualified customers use before buying more traffic to the same broken paths.

Replacing metric overload with an evidence-led recovery system

The work changed the business question from “Why are sales down?” to a set of answerable decisions about acquisition, experience, conversion, customer value, and recoverable infrastructure.

THREE DIAGNOSTIC CONCLUSIONS

1 / TRAFFIC LOSS DOMINATED

Safari Mobile traffic fell 60.2%. The 41.2% relative conversion gain could not offset the smaller audience.

2 / EXPERIENCE IMPROVED

Real-user Core Web Vitals moved from failing to fully green, reducing the evidence for speed as the sole cause.

3 / RECOVERY WAS STRUCTURAL

Search visibility, product relevance, feed health, retention, and qualified traffic offered higher-leverage paths under constraint.

DECISION-MAKING VALUE

ROOT-CAUSE CLARITY

Separated channel loss from conversion and technical performance.

REVENUE MODELING

Quantified a conservative + $27.9K monthly Safari Mobile opportunity.

RESOURCE FOCUS

Redirected limited capacity toward recoverable structural levers.

MEASUREMENT DISCIPLINE

Kept actuals, estimates, and recommended work visibly distinct.

CROSS-FUNCTIONAL TRANSLATION

Connected analytics to product, UX, SEO, lifecycle, and operations.

STRATEGIC TAKEAWAY

The strongest recovery opportunity was not a single redesign or campaign. It was a coordinated system for restoring qualified visibility while preserving an experience that had already become more effective.

MY OWNERSHIP

ANALYSIS + STRATEGY

Diagnostic framing · segmentation · Core Web Vitals interpretation · revenue scenario modeling · search/feed investigation · retention and audience analysis · prioritization

CAPABILITIES DEMONSTRATED

Behavioral analytics · mobile-first strategy · product and growth leadership · decision modeling · cross-channel systems thinking · executive communication